Programming with Applications in Electronic Engineering - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Programming with Applications in Electronic Engineering - NUIST (A35619)

Short Title:	Programming with Applications (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Software and Information Systems		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

The aim of this module is to highlight the computational approach to solving electrical engineering problems. The concepts and techniques covered in this module will be applicable in projects or problem-solving work that contain a strong computational with particular emphasis on numerical methods for computation. The module is designed such that a significant fraction of the student's time is spent programming/solving specific engineering problems, rather than learning abstract techniques.

Programmes

`MSTU-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / E
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	1 / 2 / E
`MSTU-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / E

Indicative Content

- Introduction to scientific programming: floating point numbers and error analysis, machine precision, general classification of computational problems, procedural programming constructs such as sequential, conditional and looping constructs.
- Overview of computational problems: integration and differentiation, interpolation and extrapolation, ordinary differential equations, partial differential equations, matrix methods, curve fitting, Monte Carlo other simulation methods
- Scientific programming tools: Matlab (or equivalent language such as Octave or Scilab), Python (or an equivalent language, such as Julia or Java), LaTeX, Markdown, etc.
- Programming applications: Periodigrams, windowing function, code implementation of digital signals, convolution -- FIR filters, signal sequences, time varying FT (STFT), and wavelets.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Utilise the procedural programming paradigm for problems in electronic engineering, such as code implementation of digital signals, periodigrams etc.
2. Design algorithms and programs using pseuduocode and flowcharts.
3. Demonstrate proficiency with linear algebra within Matlab(or similar language).
4. Explore aspects of statistics and probability such pseudorandom number generators, simple stochastic models and inclusion of noise.
5. Utilise essential methods and techniques (error analysis, problem classification) for symbolic/ numerical computation in engineering.

Learning and Teaching Methods

- The module is covered through a series of practical problem solving sessions and directed independent learning.
- Due to the very practical nature of the skills to be acquired in this module, these practical sessions will be centred around the idea of learning by doing, whereby students develop proficiency in the specified skill set through guided activities, and whereby lecturers provide short formal presentations of relevant concepts and technologies, as well as practical tips, feedback, and advice on best practices.
- Active engagement with frequent practice on examples is strongly encouraged through regular course work consisting of a mixture of practical activities graded from guided to self-directed tasks.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	25%	1,2
Assignment	25%	3
Practical	50%	4,5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	36	
Independent Learning	99	

Supplementary Material

- Rauscher, C., V. Janssen and R. Minihold. _Fundamentals of Spectrum Analysis_. Berlin: Rohde & Schwarz GmbH, 2007.
- Siauw, T. and A. Bayen. _An Introduction to MATLAB® Programming and Numerical Methods for Engineers_. London: Academic Press, 2014.

- Taylor, F.J. and J. Mellot. _Hands-On Digital Signal Processing_. Canada: McGraw-Hill Professional Publishing, 1998.

Requested Resources

- Room Type: Computer Lab